

Gradient Decoupling for Inductive Temporal Link Prediction: A Controlled Study of Stateful Representations

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ABSTRACT

Inductive temporal link prediction requires a model to score interactions involving entities absent from earlier observations. Jointly optimizing a temporal representation and its link readout is a natural design choice, but it need not be the most effective choice for a particular inductive protocol. This study investigates gradient routing in SR-GNN, a temporal model combining event encoding, recurrent node memory, deterministic pair-history signals and a hierarchical readout. We specify the scored gradient boundary, distinguish parameter learning from event-state evolution, and compare coupled and decoupled versions of the same registered readout. A separate freeze-then-probe procedure examines a different training trajectory. Experiments use chronological partitions of CoEdit, Wikipedia and MOOC and a fixed random-destination evaluation procedure. Checksum-bound results are reported as mean inductive average precision and sample standard deviation across the registered seeds. The observed decoupled arm has higher mean inductive average precision than the coupled arm on each corpus. This supports a bounded within-protocol comparison rather than a general claim that end-to-end learning is harmful. The analysis identifies important interpretation limits: CoEdit shares its source with Wikipedia, its event construction includes retrospective information, the candidate sampler is not destination-type constrained, and temporal memory follows mixed batch and sequential semantics. Historical experiments are reconciled at the level of configuration identity but are not pooled with the current results. The study contributes an explicit account of gradient routing and temporal evaluation that makes the observed comparison reproducible and its scope assessable.

Key words: temporal graphs, inductive link prediction, gradient decoupling, recurrent memory, reproducible evaluation

I. Introduction

Temporal graphs describe relationships through ordered interactions rather than a single static adjacency matrix. An edit on a collaborative platform or an action in an online course supplies information about both the participating entities and their recent activity. Predicting an interaction therefore involves a representation of evolving history, a candidate-scoring rule, and a decision about which historical information is available at the prediction time. These components jointly define the learning problem. A change to event ordering or candidate construction can alter the meaning of a reported score even when the neural architecture is unchanged.

Inductive prediction adds another constraint: some evaluated interactions involve entities absent from the training population. A model cannot rely solely on well-trained representations of those entities. Shared encoders, temporal neighborhoods, pair statistics and initialization rules become relevant to transfer. Nevertheless, inductive evaluation is not a unique protocol. The treatment of validation entities, the handling of preceding test events and the membership of negative-candidate pools all determine what a new entity means operationally. We state these choices explicitly instead of using the word inductive as a sufficient description of the experiment.

Most task-trained representations allow prediction gradients to update the encoder that produces their features. This gives the encoder a direct way to preserve information useful for the training objective. It can also create a dependence on the training population and on the particular candidate-sampling distribution. Stopping a scored gradient changes this optimization pressure while retaining the representation’s forward value. The resulting tradeoff is empirical: reducing task adaptation can either preserve transferable information or remove a useful learning signal. Neither outcome follows from the stop-gradient operation alone.

Temporal methods make this question especially interesting because they combine learned weights and evolving state. A detached representation is not necessarily static. Node memories can change during forward execution, pair statistics can accumulate observations, and auxiliary objectives can update parameters outside the detached scoring path. Calling the entire model frozen would conflate these mechanisms. Conversely, freezing an encoder after pretraining does not recreate the trajectory obtained by preventing a scored gradient from reaching it throughout initial training.

We study this distinction in SR-GNN, the implementation provided by the accompanying research repository [13]. Its temporal representation combines an event encoder, recurrent node memory and a bank of pair-history signals. A hierarchical readout produces the scored link output. The primary experiment changes whether that readout receives detached features while preserving the registered readout configuration. A sepa-

rate freeze-then-probe arm uses pretraining followed by a newly initialized prediction head. These are deliberately interpreted as different experimental procedures.

We ask whether the same temporal readout generalizes differently when its scored features are connected to or detached from the backbone during training. To interpret that comparison, we specify the derivative boundary and the event-state updates separately. The experiments then compare the registered profiles using matched evaluation conventions. The resulting claim concerns this intervention in SR-GNN; extending it to other architectures or candidate populations requires further experiments.

II. Related Work

1. Structural representations and link readouts

Early graph embeddings learn a vector for each observed vertex. DeepWalk derives training contexts from truncated random walks [48]; node2vec biases those walks to change the neighborhoods preserved by an embedding [22]; LINE preserves proximity through an edge-based objective [54]. These approaches establish useful structural representations, but a learned vector lookup alone does not specify how an unseen endpoint should be represented. GraphSAGE addresses that problem by learning a shared neighborhood aggregation function [24]. This distinction between stored identity and a shared mapping motivates our focus on interactions involving previously unseen nodes.

Graph convolutional networks combine local structure with node features [37], while graph attention networks learn weights for neighboring features [61]. Message-passing neural networks provide a common description of neighborhood computation [18]. The expressivity analysis of Xu et al. shows that aggregation choices constrain which structures such models can distinguish [67]. These are forward-representation choices. Our intervention instead changes the backward path at a fixed scored boundary, so a favorable routing result does not imply greater structural expressivity.

Link prediction also depends on the decoder and the information presented to it. A variational graph autoencoder combines a latent-variable encoder with a link decoder [38]; SEAL learns from subgraphs extracted around candidate links [71]. The former emphasizes latent node representations, whereas the latter makes a candidate’s local structure explicit. SR-GNN similarly supplies pair-history features to its readout, but on an ordered event stream. This creates a reason to examine whether link supervision should reshape the learned representation when candidate-level context already carries predictive information.

2. Representing change over time

Temporal-network analysis treats the ordering of contacts as part of the network, because an aggregated graph can admit paths that no time-respecting interaction sequence permits [28]. The survey

by Kazemi et al. distinguishes dynamic graph encoders and decoders across changing nodes, attributes and edges [33]. A central modeling choice is whether change is represented through snapshots or through individual timestamped events.

Among snapshot approaches, DynGEM incrementally adapts autoencoder-based embeddings [19], dyngraph2vec models transitions with dense and recurrent components [20], and dynnode2vec updates walk-based representations as the graph evolves [45]. DySAT applies attention to structural neighborhoods and temporal sequences [52]. EvolveGCN evolves graph-convolution parameters rather than maintaining a recurrent embedding trajectory for every node [46]. These methods preserve different forms of continuity; their state cannot be equated simply because each produces a time-indexed vector.

Event-based models retain finer timing information. Lee et al. construct time-respecting walks directly from edge streams [42]. Know-Evolve combines evolving entity representations with a point process for relational events [57], and DyRep jointly models dynamic representation learning and event occurrence [58]. JODIE learns co-evolving user and item trajectories [41]. Temporal Graph Networks organize event processing around memory, messages and temporal graph computation [51]. The recurrent gating used in such systems is related to gated recurrent sequence models [10], but an event store’s read and write schedule remains a separate implementation choice.

TGAT uses time-aware neighborhood attention [66], and causal anonymous walks encode temporally ordered structural patterns for inductive learning [64]. GraphMixer investigates simpler multilayer-perceptron components [11], while NAT constructs neighborhood-aware representations from dictionaries of recent interactions [44]. TCL uses a contrastive transformer formulation [63]; DyGFormer models interaction histories and accompanies the DyGLib evaluation library [69]. The Transformer supplies the underlying attention-based sequence-modeling framework [60]. These alternatives show why a routing intervention must be defined within each architecture: replacing a memory model by an attention model changes much more than its gradient boundary.

3. Objectives, probes and gradient routing

A representation can be evaluated through a separately trained classifier, as in the probes of Alain and Bengio [1]. Kumar et al. show that fine-tuning can distort pretrained features in settings where a linear probe generalizes better out of distribution [40]. Their result motivates studying the cost of task adaptation, but their pretraining assumptions differ from the stateful temporal procedure here. Our freeze-then-probe arm also retraining a different head, so it answers a different question from the shared-readout detach comparison.

Self-supervised learning offers several ways to shape an encoder without the downstream prediction loss. Contrastive Pre-

dictive Coding predicts future latent information using a contrastive objective [59]. SimCLR compares augmented views within a batch [8], and MoCo maintains a momentum-based representation dictionary [26]. Supervised contrastive learning incorporates class labels into the comparison objective [34]. These methods change the learning target and the comparison population. They should not be described as equivalent to removing a derivative from a supervised link score.

BYOL learns by predicting a target representation [21], SimSiam examines a simpler Siamese design with stop-gradient [9], and DINO studies self-distillation in vision transformers [6]. Barlow Twins instead constrains cross-correlation between views [70], while VICReg explicitly regulates variance and covariance along with agreement [4]. Their different ways of avoiding representation collapse make the computational role of gradient asymmetry relevant. They do not supply a general theorem that detached temporal features should outperform jointly trained ones.

Graph-specific objectives further separate representation learning from the downstream readout. Deep Graph Infomax relates local representations to graph summaries [62]; GraphCL studies graph augmentations [68]; GRACE contrasts node representations across perturbed graph views [72]. BGRL bootstraps alternative graph views without requiring contrastive negatives [55], and GraphMAE reconstructs masked node features [29]. These are possible sources of alternative pretraining for a future probe study. We do not benchmark them in the present matrix or infer their temporal behavior from static-graph evaluations.

The information bottleneck formulates a tradeoff between retained input information and predictive information [56]. Its deep variational formulation [2] and variational autoencoding [36] motivate tractable regularization objectives. However, the bottleneck functional can have problematic behavior for deterministic neural representations [3], and variational mutual-information bounds have estimator-dependent bias and variance [49]. A KL penalty in SR-GNN therefore specifies an optimization term, not a measured removal of identity information. This is also distinct from dropout’s suppression of feature co-adaptation [27] and from decoupled weight decay in adaptive optimization [43].

4. Evaluation and transfer beyond the benchmark

Poursafaei et al. show that dynamic link-prediction evaluation depends strongly on candidate sampling and introduce EdgeBank as a memorization-based comparator [50]. TGB provides standardized temporal benchmarks [31], and TGB 2.0 extends evaluation to temporal knowledge graphs and heterogeneous graphs [16]. OGB addresses comparable dataset and evaluator conventions for graph learning more broadly [30]. These efforts support specifying endpoint roles, historical support and split boundaries alongside an accuracy measure.

Fair comparisons also require aligned training and selection procedures. Shchur et al. document sensitivity of graph-model

rankings to evaluation choices [53]. Dacrema et al. identify weaknesses in reproducibility and baseline tuning in neural recommendation research [14]. Cawley and Talbot analyze selection bias from optimizing a finite-sample model-selection criterion [7], while Bouthillier et al. examine multiple sources of benchmark variance [5]. Accordingly, our seed deviations describe the selected runs; they do not capture uncertainty over new datasets or a repeated hyperparameter search.

Transfer raises questions that a held-out score alone cannot settle. Shortcut learning can exploit signals that fail under a harder test distribution [17]. Underspecification can leave models with similar held-out performance but different deployment behavior [12], and WILDS demonstrates the value of evaluating naturally occurring distribution shifts [39]. Leakage analysis further requires matching the information available to the claimed prediction setting [32]. These concerns are directly relevant to CoEdit’s midpoint timestamps and to the endpoint roles of sampled negatives. Finally, ranking quality and probability calibration are different properties [23]; average precision does not establish calibrated interaction probabilities.

5. Online history and implementation

Hawkes processes model self-exciting event behavior [25], and Welford’s recurrence updates running moments without retaining the full sample [65]. SR-GNN uses intensity-like and moment channels inspired by these ideas. Its supplied gap is source-node staleness, however, so the channels are not a fitted pair-specific generative point process. Their implemented semantics are specified below.

PyTorch provides the automatic differentiation and stateful modules used by the implementation [47]. PyTorch Geometric illustrates how graph libraries package sparse operations and batching conventions [15]. Such libraries assist implementation, but do not by themselves define the historical information available at scoring time. The distinction matters here because deterministic pair accumulators and recurrent node memory follow different within-batch update schedules.

III. Problem and Model Specification

1. Events, candidates and information

Let an event stream contain tuples $e_i = (u_i, v_i, t_i, x_i)$ in non-decreasing timestamp order. The vector x_i contains attributes supplied by the data pipeline. Let $\mathcal{H}_i$ denote the history retained by the particular training or evaluation pass immediately before scoring event i . It need not equal the full raw stream prefix: filtering an evaluation stream before execution changes the observations that enter its state. The model assigns a score to a candidate pair conditioned on this retained history and on the supplied attributes.

We write the temporal representation and output as

$$\begin{aligned} h_i &= F_\theta(u_i, v_i, t_i, x_i; \mathcal{H}_i), \\ p_i &= \sigma(G_\psi(h_i, \phi_i)), \end{aligned} \quad (1)$$

where θ and ψ denote backbone and readout parameters, respectively, and ϕ_i denotes the available pair-history context. This decomposition names the scored boundary; it is not an assertion that every auxiliary quantity is independent of every parameter in F_θ . In the implementation, candidate-specific positive and negative branches have corresponding representations and histories.

2. Deterministic pair-history updates

For an ordered pair, let g_k be the gap supplied to its deterministic update at its k th retained observation. Denote the running mean by μ_k and the sum of squared deviations by M_k . The online moment recurrence is

$$\begin{aligned} \delta_k &= g_k - \mu_{k-1}, \\ \mu_k &= \mu_{k-1} + \delta_k/k, \\ M_k &= M_{k-1} + \delta_k(g_k - \mu_k). \end{aligned} \quad (2)$$

The feature variance is formed from the accumulated moments using the implementation’s population-moment convention. This within-pair feature variance must not be confused with the sample standard deviation across independent training seeds used in the result table. The two quantities answer different questions and have different denominators.

In the current forward implementation, the supplied gap is source-node staleness from the node-memory store. It is not necessarily the elapsed time since the same pair last interacted. The statistics are indexed by pair but accumulate this supplied gap. This distinction limits a literal interpretation as inter-arrival moments of a pair-specific point process.

The recurrence and intensity-like updates can be expressed as

$$r_k = a + (1 - a)r_{k-1} \exp(-\eta g_k), \quad (3)$$

$$\ell_k = \mu_0 + (\ell_{k-1} - \mu_0) \exp(-\beta g_k) + \alpha. \quad (4)$$

Here r_k summarizes recent recurrence, ℓ_k is the intensity-like channel, and the constants govern decay and event increments. Equation (4) includes the baseline and event increment. Omitting either would describe a different update law. The resulting quantity is used as model context rather than reported as a calibrated event probability.

Fast and slow exponentially weighted rate estimates summarize changes in cadence. With $q_k = 1/\max(g_k, \epsilon)$, their generic form is

$$\begin{aligned} f_k &= a_f q_k + (1 - a_f)f_{k-1}, \\ s_k &= a_s q_k + (1 - a_s)s_{k-1}. \end{aligned} \quad (5)$$

The gap floor keeps coincident observations finite. Relative fast and slow rates and a decaying rate peak provide additional context for the readout. Their values depend on time units and update ordering. These conventions must therefore travel with a result rather than being inferred from a feature name.

The model also maintains learned node-memory and pair-representation components. Deterministic statistics offer a direct summary of observations, whereas learned transformations can adapt how the summaries and event attributes are combined. Decoupling the scored path changes the latter adaptation without deleting the former information. This is why an effective detached arm does not imply that a random or history-free encoder would be equally effective.

3. Readout and gradient boundary

Let $D_0(h) = h$ and $D_1(h) = \text{sg}(h)$, where stop-gradient is defined by its forward value and backward derivative:

$$\text{sg}(h) = h, \quad \frac{\partial \text{sg}(h)}{\partial h} = 0. \quad (6)$$

The primary arms use the same registered readout family and differ in whether its scored feature input passes through D_0 or D_1 :

$$p_i^{(d)} = \sigma(G_\psi(D_d(h_i), \phi_i)), \quad d \in \{0, 1\}. \quad (7)$$

This boundary is applied to both positive and negative scoring branches. Detaching only observed interactions would leave a task-gradient route through the sampled negatives and would implement a different intervention.

For the path through h_i , the chain rule gives

$$\nabla_{\theta} L_{\text{score}}|_{h_i} = \frac{\partial L_{\text{score}}}{\partial G} \frac{\partial G}{\partial D_d} \frac{\partial D_d}{\partial h_i} \frac{\partial h_i}{\partial \theta}. \quad (8)$$

That contribution vanishes for the detached branch. This is a path-specific computational statement. It does not establish that every route from the complete weighted training objective to every backbone parameter vanishes. Additional dependencies through weights, shared modules or auxiliary losses must be inspected separately. In particular, the current soft-compliance weighting should not be silently treated as a detached constant when analyzing the entire objective.

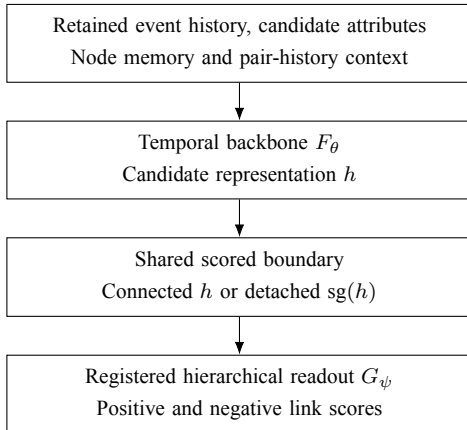


Figure 1. Scored feature path in the primary comparison.

Figure 1 identifies the boundary changed by the experiment. Auxiliary objectives and deterministic store writes are not drawn as scoring edges. Their omission from this diagram is a visual simplification, not a claim that they are absent from training. The forward representations of arms with identical parameters would agree at the boundary, but their learned trajectories need not agree after optimization.

4. Training objective and auxiliary routes

For a scored logit z and label y , the binary logistic loss is

$$\ell(z, y) = -y \log \sigma(z) - (1 - y) \log(1 - \sigma(z)). \quad (9)$$

The implementation combines weighted per-example prediction losses with enabled auxiliary terms. A compact description is

$$L = \frac{1}{|\mathcal{B}|} \sum_{j \in \mathcal{B}} w_j \ell(z_j, y_j) + \lambda_{\text{KL}} L_{\text{KL}} + \sum_{a \in \mathcal{A}} \lambda_a L_a. \quad (10)$$

The auxiliary collection includes the active violation, transition, edge-state and entropy-related terms selected by the resolved profile. Disabled terms contribute nothing. The code, protocol and resolved configuration specify their actual inclusion and weights. Equation (10) is a dependency summary rather than a substitute for that operational specification.

5. State updates and batching

Scoring and store mutation must be ordered consistently for positive and negative candidates. A positive candidate reads pre-update pair history; a negative candidate reads its own available pair history without inserting a fictional observation. Actual observations subsequently update the stores. If positives instead received their post-event counts while negatives saw only pre-event counts, the model could exploit an evaluation artifact rather than historical predictive information.

Repeated pairs within a batch require additional care. A single batch snapshot would expose the same old pair statistics to all repeated occurrences. The causal-batch option folds deterministic pair accumulators in stable event order so a later occurrence can read the state following an earlier occurrence in that batch. This addresses a store-level timing issue. It should not be interpreted as sequential execution of the entire neural model.

Recurrent node memory follows a different registered convention. Candidate updates are computed from a batch snapshot, and collisions on a node index are resolved by committing the final candidate in stable input order. Thus the procedure combines sequential deterministic pair updates with batch-snapshot neural memory. This is a reproducible approximation whose batch size is part of the protocol. A comparison to fully event-sequential neural updates would require a separate experiment.

6. Freeze-then-probe procedure

The freeze-then-probe arm pretrains with the separate main prediction head, selects a validation checkpoint, freezes representation parameters and reinitializes the main head. The probe phase then trains the new head while forward event processing continues to update temporal stores. Its own validation selection determines the evaluated checkpoint. Freezing weights therefore differs from freezing a precomputed feature matrix.

IV. Data and Experimental Protocol

1. Corpus construction and identities

Wikipedia and MOOC are obtained from the public JODIE data sources [41]. They are bipartite streams whose endpoint roles have separate node-ID namespaces in the current preprocessing. The distinction prevents a user and an item with the same original integer label from sharing a memory row. CoEdit instead represents contributor-to-contributor interactions in a homogeneous namespace. Topology is a property of each registered corpus and is not inferred from whether source and destination integers happen to overlap.

CoEdit is derived from Wikipedia edits. For each page, the builder considers edits by distinct contributors within a 60-minute window. It emits a contributor pair with the average of the edit attributes and the midpoint of their timestamps. It retains the first 80,000 qualifying pairs in page first-observation order and then applies stable chronological ordering. This procedure is not equivalent to taking the earliest 80,000 qualifying pairs globally. The selection order may influence which pages and recurrence patterns are represented.

The midpoint convention also limits a forecasting interpretation. The later edit is required to construct a co-edit event, yet its attributes enter a derived event timestamped before that later edit. Thus chronological ordering of the derived stream does not make all its attributes prospectively available at the midpoint. The present experiment is a comparison on this registered derived corpus. A deployment-oriented forecasting study should timestamp a constructed interaction at its actual availability time or exclude attributes unavailable at the proposed decision time.

CoEdit and Wikipedia consequently do not provide independent source replications. They offer different graph constructions from a shared source, whereas MOOC supplies a separate interaction source. This distinction matters when assessing how broadly an observed ordering generalizes. An effect repeated across these corpora is more informative than an effect on a single graph construction, but it is not evidence from an equal number of independent data-generating environments.

2. Chronological partitions and inductive population

The registered split assigns the chronological event stream to training, validation and testing in proportions 70/15/15. The in-

ductive node set contains nodes occurring in the test partition but in neither training nor validation. An inductive test event touches at least one such node. This definition makes the novelty boundary later than a definition that treats every node absent from training as unseen.

The evaluator filters the test events before executing the inductive pass. As a result, a non-inductive test event omitted by that filter does not update the inductive evaluator’s memory, even if it precedes a retained event in the raw stream. Training and validation history are replayed to initialize the evaluation state. This filtered-stream convention should be distinguished from replaying every test event and calculating the metric only on an inductive subset. The latter exposes the model to a different retained history.

New-node initialization is part of the model behavior rather than a separate source of supervision. Shared functions can score a candidate with a new endpoint, but the memory available for that endpoint depends on which prior retained events touched it. An interaction involving an inductive node is not necessarily its first evaluated interaction. Repeated appearances within the test period can therefore contribute useful history even in the inductive metric.

3. Candidate sampling

For each observed interaction the evaluator constructs a sampled negative while retaining the source. The destination pool follows the seen or unseen status of the true destination. A sampled destination identical to the true one is rejected. This does not amount to rejecting every true interaction at the same timestamp, and it does not ensure that the replacement endpoint has the destination role required by a bipartite application.

The same evaluation sampling convention is used across primary arms, with reset evaluation randomness supporting matched draws. This makes the internal contrast more interpretable than a comparison in which each arm sees different candidates. It still leaves a limitation shared by all arms: a candidate pool can be easier than the application population. Near-ceiling performance must be interpreted in that context rather than as evidence that arbitrary future links are predictable.

Negative candidates receive their own historical pair context and do not write themselves into memory as observed events. Reusing the positive pair’s history for a negative would erase the candidate distinction, while forcing every negative history to be empty would introduce another asymmetry. The implemented read-only lookup avoids these specific mistakes. It does not settle all potential leakage or candidate-validity questions.

4. Optimization and model selection

Table 1 summarizes the registered training configuration. The primary arms use the same backbone size, batch size, optimizer family and schedule. Adam is used for optimization [35]. The

registered optimizer uses its weight-decay setting; it is not replaced by AdamW, whose decoupled decay changes the update rule [43]. Validation-based model selection takes place within the registered training procedure; the test endpoint is not used to select an epoch. The freeze-then-probe profile includes an additional probe phase and must not be described as having the same optimization trajectory as the primary arms.

Table 1. Registered experimental settings.

Setting	Registered value
Corpora	CoEdit, Wikipedia, MOOC
Seeds	1, 7, 42
Chronological split	70/15/15
Training epochs	20
Probe epochs	20 after pretraining
Hidden dimension	128
Batch size	500
Learning rate	0.001
Weight decay	0.00001
Optimizer and schedule	Adam; cosine annealing
Primary readout	Shared hierarchical readout
Scored feature routing	Connected or detached
Compliance	Soft; floor 0.05
Negative regime	Fair random destination

The runtime warmup is state-neutral. Although a real training pass initializes the computational kernels, parameters, buffers, optimizer state, random streams and temporal stores are restored afterwards. This prevents initialization work from becoming hidden extra optimization. The runner also checks finite values and records failures rather than replacing invalid values or silently dropping failed tasks from an aggregate.

5. Metric and uncertainty

Average precision summarizes precision at increasing recall levels on the evaluated positive and negative scores. For recall thresholds indexed by k , it can be written as

$$AP = \sum_k (R_k - R_{k-1}) P_k, \quad (11)$$

where P_k and R_k are precision and recall at the corresponding threshold. The experiment uses the reference implementation recorded by the runner. Average precision is conditional on the sampled candidates and their label balance; it is not a calibrated probability that a predicted interaction will occur in deployment.

For selected seed values $a_1, \dots, a_n$, the reporting convention is

$$\bar{a} = \frac{1}{n} \sum_{j=1}^n a_j, \quad s = \sqrt{\frac{\sum_{j=1}^n (a_j - \bar{a})^2}{n-1}}. \quad (12)$$

The reported standard deviation describes variation across the retained training seeds. It is not a confidence interval for performance on new graphs, an uncertainty interval over all possible

test populations, or a measure of temporal dependence between individual test events. We do not infer significance from overlap or non-overlap of these descriptive deviations.

A paired effect would be computed from matched per-seed differences, not by subtracting standard deviations from separate arms. Establishing a confidence interval for such an effect requires its own statistical specification and sufficient supporting data. The principal statement here is the observed ordering of the registered means. It does not require or imply a claim that every seed favors the same arm.

V. Results

1. Current within-protocol comparison

Table 2 reports the current registered inductive average precision. Values are rounded to four decimal places from the checksum-bound matrix. Every row contains the same registered seed count. The connected and detached profiles use the shared scored readout configuration, so the primary comparison is not a replacement of that readout by a different predictor.

Table 2. Inductive AP: mean and sample SD.

Corpus	Profile	AP
CoEdit	Coupled	0.9787 ± 0.0021
	Decoupled	0.9891 ± 0.0016
	Freeze then probe	0.8381 ± 0.0484
MOOC	Coupled	0.9910 ± 0.0047
	Decoupled	0.9947 ± 0.0007
	Freeze then probe	0.9862 ± 0.0009
Wikipedia	Coupled	0.9967 ± 0.0003
	Decoupled	0.9983 ± 0.0003
	Freeze then probe	0.9623 ± 0.0128

Under the registered current-corpus protocol, the observed decoupled SR-GNN arm has higher mean inductive average precision than the coupled end-to-end arm on each registered corpus. This is a within-protocol arm comparison, not a baseline, state-of-the-art, architecture-general, or causal claim. The table also preserves the observed freeze-then-probe arm rather than replacing it with a qualitative label.

2. Interpreting the probe arm

The probe rows characterize the specified pretraining and head-retraining procedure. Their interpretation depends on which representation was pretrained, how its checkpoint was selected, which parameters were frozen and which head was reinitialized. Those factors differ from simply reconnecting the shared primary readout. The result therefore does not isolate the gradient boundary with the same scope as the primary comparison.

A probe can fail to reach another arm’s score because of its capacity, optimization schedule, initialization, objective, or the suitability of the pretrained features. These explanations are not

mutually exclusive. Establishing irreversible damage would require showing that relevant recovery interventions fail under a meaningful and adequately explored criterion. The current procedure supports no such universal conclusion.

3. Execution completeness and reconstruction

The selected scientific matrix covers 27 tasks. Its accounting retains 162 scientific attempts, including 57 failed and 48 cancelled attempts; 135 attempts are scientifically inadmissible. These accounting categories describe execution and selection history, not additional independent observations for a confidence calculation. The selected completed tasks determine the reported aggregates.

The release preserves failed and superseded attempts so that incomplete or unsuccessful work does not disappear from the experiment’s provenance. Means and sample deviations are reconstructed from the selected per-seed records. That reconstruction checks numerical consistency of the table with its underlying records. It is distinct from independently repeating training, and no new training is claimed as part of preparing this manuscript.

The recovered historical archive provides an additional audit trail for the earlier draft. It contains broader ablations, hard-negative runs, probes and proxy-model results, but not a complete current-release execution chain for each claim. Recovering those bytes is valuable because it permits configuration reconciliation. It does not justify adding their task counts to the current matrix or treating their outcomes as current confirmatory evidence.

VI. Discussion

1. What the routing contrast can explain

The mean ordering indicates that scored-feature routing is an empirically relevant design choice in this registered implementation. Several mechanisms could contribute. Restricting the scored gradient may preserve useful history features; it may reduce adaptation to the sampled training candidates; or it may change the interaction between the scored objective and auxiliary terms. The aggregate endpoint does not distinguish these explanations.

One possible explanation is adaptation to identity-related shortcuts [17]. Direct evidence would require measuring identity-related information, testing its relationship to performance, and intervening on the relevant features while holding other factors stable. A high inductive score and a favorable detached-arm ordering do not by themselves reveal which feature caused that ordering. They are outcomes to be explained, not measurements of the proposed mechanism.

The deterministic channels offer another interpretation. When recurrence and cadence already supply useful signals, preserving them may reduce the need for task-driven transformation. However, performance could also depend on learned event features and recurrent memory. A complete decomposition would

compare a deterministic-only representation, a learned representation without pair statistics, and controlled combinations under the same current protocol. Historical ablations provide design context but cannot substitute for those aligned measurements.

2. Why historical contrasts differ

The earlier development record contains several different notions of coupling. A design-preset contrast changes more than a scored boundary; a main-predictor knob can replace the readout with another head; and a detach toggle on the same head changes a more specific route. Treating all of these as repeated estimates of a single causal effect would conflate architectural and optimization changes.

The current shared-readout experiment makes the narrower comparison explicit. Both primary profiles keep the main-predictor switch off, and the detach setting distinguishes the scored representation path. This accounts for why one should not carry the earlier draft’s headline magnitude directly into the current result table. A manuscript can develop an earlier research question while correcting its intervention definition.

3. Generalization and practical use

The practical implication is to test a gradient-routing boundary when adapting this kind of model to an inductive setting. It is not to prohibit end-to-end training. A different encoder, a different candidate population or an application with weak deterministic history signals could benefit from direct task adaptation. The relevant design choice should be evaluated with the information available at the actual decision point.

For recommendation-like tasks, valid endpoint types and plausible alternatives are especially important. An evaluator drawing candidates across all node roles can reward discrimination that the deployed system never needs to perform. Replacing it with a destination-valid pool changes the task difficulty and may change the relative effect of routing. Such an evaluation should be reported as an extension with its own data and protocol identity.

For prospective forecasting, the temporal availability of event attributes is equally important. In a constructed corpus, an attribute can be attached to a timestamp even though it is known only after another event occurs. Preventing post-score memory writes does not correct that input-level availability issue. Deployment claims therefore require both a causally ordered state procedure and a causally available input construction.

4. Limitations and next experiments

The study has a limited seed budget and a restricted set of corpora and model profiles. The reported uncertainty concerns selected seed variation rather than sampling of independent environments. CoEdit’s shared origin with Wikipedia and its construction order further constrain external validity. These limits remain even

when all recorded hashes and aggregate computations agree.

The primary evaluation uses random destinations. Stronger tests should include destination-valid alternatives and clearly specified harder-negative regimes. They should preserve matched draws across arms, record pool support and avoid silent fallbacks when a requested pool is empty. A failed or degenerate pool should remain visible as an evaluation outcome rather than being interpreted as evidence of robustness.

The intervention should also be replicated within faithful implementations of other temporal architectures. A sound replication must preserve each model’s forward computation and change a well-defined gradient boundary. Replacing a published architecture by a simplified local approximation would answer a different question. Probe extensions should vary head capacity and recovery budget before proposing an explanation based on irreversibility.

Finally, a full gradient audit should examine both scored-output ancestry and the complete weighted objective. It should include candidate branches and auxiliary dependencies, and separate parameter gradients from event-state writes. Such an audit can establish a computational property for a specific configuration. It cannot by itself determine the statistical reason for an observed generalization difference.

VII. Conclusion

This study compares connected and detached scored-feature paths in a registered SR-GNN implementation for inductive temporal link prediction. The observed decoupled arm has higher mean inductive average precision on each registered corpus. The result supports evaluating gradient routing as an explicit design choice within the stated model and candidate-sampling protocol.

The fuller specification distinguishes scored derivatives, auxiliary optimization, recurrent memory and deterministic pair-history updates. It also exposes the role of filtered test history, candidate validity and attribute availability in interpreting a temporal result. The observed probe arm remains a separate training procedure, and historical configurations remain distinguishable from the current shared-readout comparison. These distinctions provide a basis for further experiments without turning a bounded performance observation into a universal or causal claim.

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歸納式時序連結預測中的梯度解耦： 具狀態表徵的受控研究

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摘要

歸納式時序連結預測需要處理先前觀測中未出現的節點。本研究以結合事件編碼、循環節點記憶、確定性節點對歷史訊號與階層讀出器的 SR-GNN，探討評分路徑的梯度是否應更新時序表徵。我們明確區分梯度邊界、參數學習與事件狀態演化，並在相同已註冊讀出器下比較耦合及解耦訓練；另以先訓練再凍結並重新訓練預測頭的方法作為不同訓練流程的對照。研究採用 CoEdit、Wikipedia 與 MOOC 的時間切分及固定隨機負樣本協定。具校驗碼的封存結果顯示，解耦組在各語料上的平均歸納式平均精確率均高於耦合組。此結果支持協定內的觀察比較，而非端到端學習普遍有害的結論。CoEdit 與 Wikipedia 共享來源，且其事件建構包含事後資訊；候選取樣未限制目的端節點類型，記憶更新亦混合批次與循序語意。歷史實驗依設定差異加以辨識，不與目前結果合併。本研究提供可重現的梯度路由及時序評估說明，並界定效能比較的適用範圍。

關鍵詞：時序圖，歸納式連結預測，梯度解耦，循環記憶，可重現評估